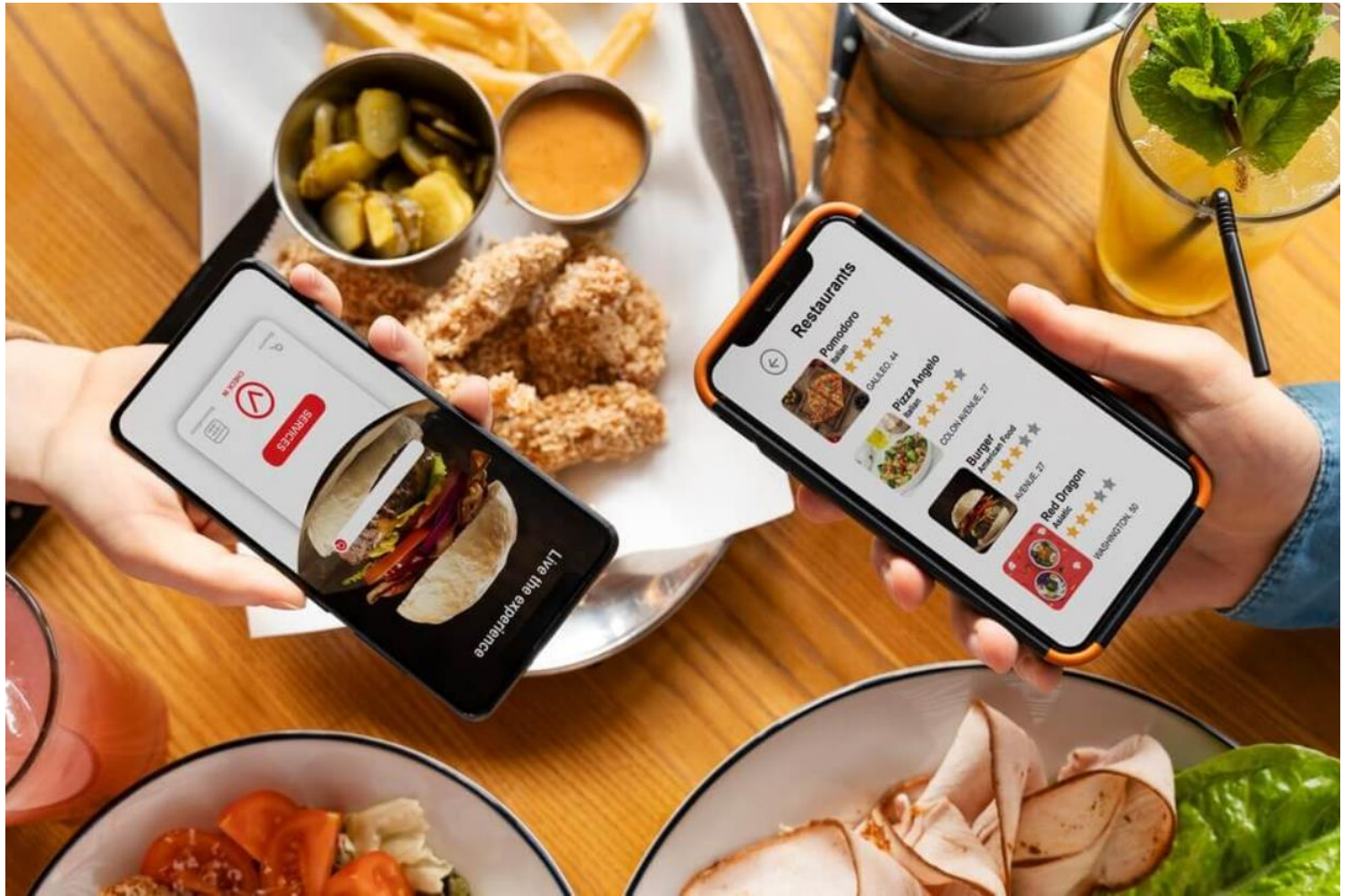




RESTAURANT APP FOR ONLINE ORDERS

Database Model
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Domain description

Restaurant app with online ordering as main option. Clients need to register user account - specifying username, password, first and last name, email address, phone number and home address. Every registered user is assigned a 'USER' role by default. There would be also 'ADMIN' role with the respective privillages - for resetting user credentials, adding/updating/de-activating products, updating promotions and etc. The main users table consists of only the details needed for authentication and authorization - username, password and role. The personal details are in separate user_info table. Each user has only one active cart, which consists of different items (with their respective price and quantity), as well as total cart_price, calculated dynamically. The products have their name, type (starters, soups, desserts and etc.), image URL, prices, promotions. They can be de-activated (but should not be deleted, as they are part of the order history). Each user can see their orders history - date and time, address (which can be different than their home address), total price and products (with their historical price at the time of the order).

DB structure

user entity

- id (auto-generated)
- username
- password
- role

user_details entity

- id (auto-generated)
- user_id - referencing id from users table (one-to-one relation)
- email
- first_name
- last_name
- phone_number
- address

carts entity

- id (auto-generated)
- user_id - referencing id from users table (one-to-one relation)
- total_price

cart_items entity

- id (auto-generated)
- cart_id - references id from carts table (many cart items-to-one cart relation)
- product_id - references id from products table (one-to-one relation)
- quantity

products entity

- id (auto-generated)
- name
- type
- image_url
- price
- is_on_promotion - boolean
- promo_price
- is_enabled - boolean, if the product is currently offered

order_history entity

- id (auto-generated)
- user_id - references id from users table (many historical orders-to-one user relation)
- total_price (historical price, by the time of the order)
- time - timestamp
- address (may not be the same as the default address of the client)

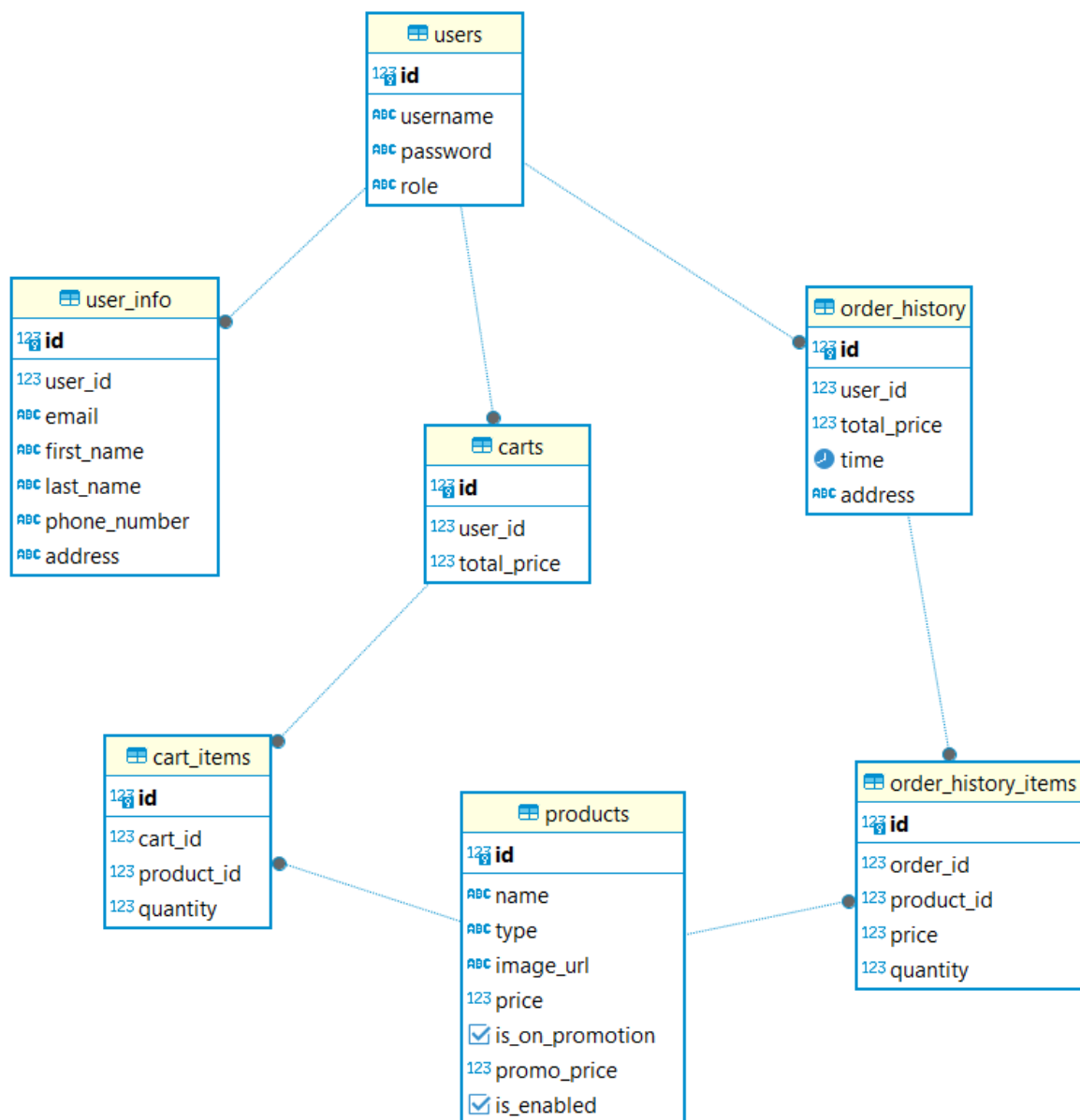
order_history_items entity

- id (auto-generated)
- order_id - references id from order_history table (many items-to-one order relation)
- product_id - references id from products table (one-to-one relation)
- price - (historical product price, by the time of the order)
- quantity

Additional information:

- each ADMIN can change usernames, reset password, reset cart contents, change product details - including enable/disable products and promotions
- ADMIN can not see user password - just update it to something else
- each USER can change own username, personal details, add and delete products to carts, change quantity, finalize order, get access to personal order history
- whenever user finishes an order - new records with the respective details are inserted into order_history and order_history_items tables. Then current cart's total_price for the user is set to 0, while the cart items related to it are deleted. Also when user removes cart_items from their cart, they are deleted from cart_items table.
- typically promo price is 20% off the regular product price
- all prices are in euro

ER diagram



SQL script for creating the database

```
CREATE TABLE users(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    username VARCHAR(30) NOT NULL UNIQUE,  
    password VARCHAR(30) NOT NULL,  
    role VARCHAR(30) NOT NULL  
);  
  
CREATE TABLE user_info(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    user_id INT NOT NULL REFERENCES users(id),  
    email VARCHAR(30) NOT NULL UNIQUE,  
    first_name VARCHAR(30) NOT NULL,  
    last_name VARCHAR(30) NOT NULL,  
    phone_number VARCHAR(30) NOT NULL UNIQUE,  
    address VARCHAR(100) NOT NULL  
);  
  
CREATE TABLE products(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    name VARCHAR(30) NOT NULL,  
    type VARCHAR(30) NOT NULL,  
    image_url VARCHAR(150) NOT NULL,  
    price DECIMAL(9, 2) NOT NULL,  
    is_on_promotion BOOLEAN NOT NULL,  
    promo_price DECIMAL(9, 2) NOT NULL,  
    is_enabled BOOLEAN NOT NULL  
);  
  
CREATE TABLE carts(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    user_id INT NOT NULL REFERENCES users(id),  
    total_price DECIMAL(9, 2) NOT NULL  
);  
  
CREATE TABLE cart_items(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    cart_id INT NOT NULL REFERENCES carts(id),  
    product_id INT NOT NULL REFERENCES products(id),  
    quantity INT NOT NULL  
);  
  
CREATE TABLE order_history(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    user_id INT NOT NULL REFERENCES users(id),  
    total_price DECIMAL(9, 2),  
    time TIMESTAMP NOT NULL,  
    address VARCHAR(100) NOT NULL  
);  
  
CREATE TABLE order_history_items(  
    id INT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
    order_id INT NOT NULL REFERENCES order_history(id),  
    product_id INT NOT NULL REFERENCES products(id),  
    price DECIMAL(9, 2) NOT NULL,  
    quantity INT NOT NULL  
);
```